

Gyanaranjan Mishra

Applied Data Scientist

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SUMMARY

From steel process engineering at Tata Steel to PhD research in materials modeling, data science has been a consistent part of my work, evolving from applied statistics and empirical modeling into end-to-end ML pipelines during my PhD. Proficient in Python, SQL, TensorFlow, XGBoost, and scikit-learn, with hands-on experience across the full workflow from data cleaning and feature engineering to model deployment and performance monitoring. Working knowledge of LLM-based tools including RAG pipelines and prompt engineering, with a focus on evaluating model and system output quality and reliability. AWS Certified Machine Learning Engineer - Associate and AWS Certified AI Practitioner. Actively looking to apply the same skill set into a dedicated Data Scientist or ML Engineer role, backed by a track record of learning fast and adapting across domains.

SKILLS

Programming and Tools: Python (Pandas, NumPy, Scikit-learn, TensorFlow, Keras), SQL (Joins, Subqueries, Window Functions, Aggregations), MATLAB, SPSS, Git, AWS (SageMaker, S3), Jupyter Notebooks

Machine Learning and Deep Learning: Regression, Classification, Clustering, Random Forest, XGBoost, Neural Networks, Deep Learning, NLP (Word2Vec, NLTK)

ML Engineering: Full-Cycle ML Pipeline Development, LLM Integration (RAG, FAISS, Reranking, Prompt Engineering), Feature Engineering, Hyperparameter Tuning, Cross-Validation, Model Serialization, Streamlit Deployment, Eval Metrics (ROC-AUC, F1, Precision-Recall)

Analytics and Visualization: EDA, Matplotlib, Seaborn, Plotly, Power BI, ETL

Domain: Hypothesis Testing, A/B Testing, Experimental Design, Statistical Quality Control, Process Optimization, Project Management

RELEVANT EXPERIENCE

Principal Researcher (Process Analytics and ML), New Product Development

Oct 2018 - Aug 2021

Research and Development, Tata Steel Ltd, India

- Built multiple linear regression and ensemble models (XGBoost and Random Forest) in Python to predict properties of new steel products across market segments during downstream processing (after rolling), replacing manual property evaluation tests and reducing QA report preparation time by nearly 40 % (from roughly 12 to 7 hours).
- Built ML models as a root cause analysis tool to identify factors driving high oxide volume during slab reheating (downstream processing), reducing yield loss from roughly 3 - 4 % to 1.9 - 2.8 %, a meaningful recovery given individual slabs weigh 2 - 3 tons.
- Partnered with production, IT, and quality teams to embed ML-driven analytics into daily operations as part of a plant-wide Industry 4.0 transformation across Tata Steel India.
- Mentored junior researchers and coordinated ML modeling adoption across production, quality, and IT teams, expanding use of predictive models beyond the original pilot unit.
- Performed internal quality audits to maintain the in-house Process Management System (PMS) for Tata Steel R&D, reviewing procedural compliance, documentation integrity, and corrective action closure across laboratory operations and developmental projects.

Researcher, New Product Development

Jul 2012 - Sep 2018

Research and Development, Tata Steel Ltd, India

- Analyzed process, quality, and metallurgical datasets received in batches from Level 2 MES, and prepared reports highlighting insights into process, quality, and property variations across steel grades, supporting data-backed decisions during new product development.
- Developed empirical models for alloy design, providing cost-effective alternatives to physical trials and accelerating the ideation-to-plant-trial cycle for new steel grades.

- Conducted multi-physics simulations (ThermoCalc, DICTRA, MatCalc, JMatPro) for alloy design, precipitation kinetics, and phase transformation modeling, using Python and Excel to visualize results and surface key insights.

DOCTORAL RESEARCH EXPERIENCE

Visiting Researcher (Applied Data Scientist), Advanced Materials Processing

Apr 2022 - Sep 2025

CanmetMATERIALS, Natural Resources Canada (unpaid, concurrent with PhD, University of New Brunswick)

- Built XGBoost, Random Forest, and Neural Network regression models on a 200K+ sample dataset to predict high-temperature material deformation behavior; outperformed the empirical physics-based model by 67 % in prediction accuracy (MAE of less than 2 % vs. 7–13 % baseline), enabling its adoption for data generation supporting concurrent research activities.
- Engineered analytics pipelines covering data ingestion, cleaning, feature engineering, and model evaluation, improving dataset reliability by 25 % (reducing data quality errors from roughly 20 % to 15 % of records) and accelerating insight delivery to industry partners.
- Deployed predictive models as interactive Streamlit applications, enabling non-technical government and industry stakeholders to query model outputs directly without engineering support.
- Translated model results into technical reports that informed strategic decisions for government and industry partners.

SELECTED PROJECTS

- **Document Q&A – RAG Pipeline** | [Link](#) - Built an end-to-end RAG system over 7,300+ text chunks from 5 books, combining MPNet embeddings, FAISS cosine-similarity retrieval, and CrossEncoder reranking to surface relevant passages; engineered prompts for an LLM (Claude Haiku) to generate source-attributed answers at ~\$0.0014 per query.
- **Deformation Stress Modeling (PhD)** | [Link](#) - Built two deep learning models for different steel groups (TensorFlow, Keras) on 200K+ multi-source samples to predict high-temperature material behavior, achieved ~67 % accuracy improvement over conventional models. Deployed as an interactive Streamlit app end-to-end from raw data to user-facing interface.
- **Quora Question Pair Similarity** | [Link](#) - Engineered an NLP pipeline combining Word2Vec embeddings (Gensim) with fuzzy matching and token overlap features to detect semantically similar question pairs. Trained a Random Forest classifier achieving ~90 % accuracy.
- **Imbalanced Classification - Kick Car Prediction** | [Link](#) - Built an end-to-end ML pipeline on 70K+ records and 30+ features with XGBoost and Random Forest classifier, handled class imbalance, adjusted threshold to increase recall, and optimized PR - AUC. Deployed as a Streamlit app with serialized model artifacts.

CERTIFICATIONS

- AWS Certified Machine Learning Engineer, Associate, Amazon Web Services Jun 2026 - Jun 2029
- AWS Certified AI Practitioner, Amazon Web Services Apr 2026 - Apr 2029
- Unsupervised Learning, Recommenders, Reinforcement Learning, DeepLearning.AI, Stanford University Oct 2023
- Supervised Machine Learning: Regression and Classification, DeepLearning.AI, Stanford University Aug 2023
- SQL (Basic), HackerRank Apr 2026

AWARDS AND ACHIEVEMENTS

- RNB Graduate Merit Award, UNB, Canada 2021 - 2025
- NBIF STEM and Social Innovation Award, UNB, Canada 2021, 2022
- Best Innovative Project Award, Tata Steel Ltd, India 2016, 2017
- Institute Silver Medal, NIT Rourkela, India 2009

EDUCATION

- PhD, Mechanical Engineering, UNB, Canada, GPA 4.2/4.3 Sep 2021 - May 2026
- MTech, Metallurgical Engineering and Materials Science, IIT Bombay, India, GPA 8.76/10 Jul 2010 - Jun 2012
- BTech, Metallurgical and Materials Engineering, NIT Rourkela, India, GPA 9.15/10 Jul 2005 - Apr 2009